HW_Assignment_3_Fall2026

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ENEL 573 Computer Networks: HW Assignment 3 (Fall 2026)
(Due on Oct 26th, at 11:59pm MST)

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Problem 1 (40 points):
Consider the following Aloha systems.
(a) A group of N users share a 56 kbps **pure Aloha** channel. Each user generates at a Poisson rate of one 1000-bit packet every 100 sec, even if the previous one has not yet been sent. What is the maximum value of N (Hint: The pure Aloha maximum channel utilization is 18.4%)?
(b) Ten thousand airline reservation stations are competing for the use of a single **slotted Aloha** channel. The average station makes 18 requests/hour. A slot is 125 μsec. What is the approximate total channel load (requests per slot)?

a) $X = \frac{L}{R} = \frac{1000}{56\times10^3} = 17.86\,ms$

$S = N\lambda X = N\left(\frac{1}{100}\right)(17.86\times10^{-3}) = (0.1786\times10^{-3})N$

$S_{max} = \frac{1}{2e} = 0.184$

$S \le S_{max}$

$(0.1786\times10^{-3})N \le 0.184$

$N \le 1030.06$

$N_{max} = 1030$ users

b) $18\ req/hr = 5\times10^{-3}\ req/s$

$\lambda = (10000)(5\times10^{-3}) = 50\ req/s$

$G = \lambda X = (50)(125\times10^{-6}) = 6.25\times10^{-3}\ req/slot$

Problem 2 (30 points):

A large population of **slotted Aloha** users manages to generate 50 requests/sec, including both new requests and retransmissions. Time is slotted in units of 40 *m*sec.

(a) What is the chance of success on the first attempt?

(b) What is the probability of exactly k collisions and then a success?

(c) What is the expected number of transmission attempts needed to have a successful transmission (Hint: Average number of transmission attempts over throughput)?

$X = 0.04\ s/slot$

$\lambda = 50\ req/s$

$G = \lambda X = (50)(0.04) = 2\ req/slot$

a) $P(k \text{ arrivals in } X) = \frac{G^k}{k!} e^{-G} = \frac{2^0}{0!} e^{-2} = 0.135$

b) $P(k \text{ collisions, then success}) = (1 - e^{-G})^k e^{-G} = (1 - e^{-2})^k e^{-2}$

c) $E[\text{attempts}] = \frac{G}{S} = \frac{G}{Ge^{-G}} = e^G = e^2$

Problem 3 (30 points):

Suppose three active nodes, nodes A, B, and C, are competing for access to a channel using slotted ALOHA. Assume each node has an infinite number of packets to send. Each node attempts to transmit in each slot with probability p. The first slot is numbered slot 1, the second slot is numbered slot 2, and so on.

a. What is the probability that node A succeeds for the first time in slot 4?

b. What is the probability that some node (either A, B, or C) succeeds in slot 2?

c. What is the probability that the first success occurs in slot 4?

a) $P[\text{A succeeds in a slot}] = p(1-p)^2$

$P[\text{A fails in 1,2,3 and succeeds in 4}] = \left[1 - p(1-p)^2\right]^3 p(1-p)^2$

b) $P[A \cup B \cup C \text{ succeeds in a slot}] = 3p(1-p)^2$

c) $P[A \cup B \cup C \text{ fails in 1,2,3 and succeeds in 4}] = \left[1 - 3p(1-p)^2\right]^3 3p(1-p)^2$